Analisis Kemiripan Teks Menggunakan TF-IDF dan Cosine Similarity

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Deskripsi data

Terdapat 3 buah teks abstrak yang di ambil dari jurnal di link jurnal.ugm.ac.id/ijccs

Teks 1 = Report from the WHO that one of the highest causes of medication errors is Look Alike – Sound Alike (LASA) drugs, leading to errors in receiving information about the drugs, which of course will affect patient safety. Efforts to reduce medication errors have been widely implemented, such as conducting medication training, managing medications, and storing and labeling medications. However, all of that leads to human error, so the utilization of technology is needed to address this issue. The technology expected to help reduce medication errors is the utilization of artificial intelligence (AI). AI is designed for automation processes and systems that can learn independently, allowing the causes of medication errors such as LASA to be learned by the system and predicted automatically. Deep learning is a part of AI that works by providing solutions accurately and automatically. The Recurrent Neural Networks (RNN) algorithm is one of the deep learning methods that has been proven accurate in predictions based on previous research studies. In this study, LASA predictions were made using RNN with the aim of serving as an aid to reduce medication errors, thereby ensuring patient safety. The accuracy achieved is 99% for training and 81% for testing.

Teks 2 = In this study, we conducted a comprehensive sentiment analysis of users on the social media platform X concerning the viral controversy surrounding the KasKus account known as “Fufufafa.” This issue attracted widespread attention and sparked varied reactions within the online community. To gain insights into public opinion on the topic, we utilized the Support Vector Machine (SVM) method, a widely recognized machine learning algorithm for classification tasks. The data for this research was gathered from various posts, comments, and public discussions on platform X, which were pre-processed to filter out irrelevant information, such as spam, unrelated topics, and non-informative content. After cleaning the data, user sentiments were categorized into three primary classes: positive, negative, and neutral. The SVM model was then trained and tested using a labeled dataset to accurately predict user sentiments based on the textual content of their interactions.

Teks 3 = The integration of Artificial Intelligence (AI) into business operations has garnered significant attention due to its potential impact on business performance. However, the relationship between AI adoption and business performance remains not fully understood. This article comprehensively analyzes this relationship through three key aspects: the acceptance

and implementation of AI within organizations, the impact of AI on various dimensions of business performance, and the potential challenges associated with AI adoption. In this study, we employ SmartPLS as an analytical tool to evaluate the relationships between identified factors and the impact of AI adoption on business performance. Our findings reveal that several factors influence the adoption and implementation of AI, including data availability, organizational culture, leadership support, technical expertise, and ethical considerations. Moreover, AI adoption significantly influences business performance metrics such as productivity, efficiency, revenue, and customer satisfaction. Nonetheless, challenges arising from AI adoption, including shifts in job roles, data privacy, and security concerns, also require substantial attention. In conclusion, successful AI adoption and implementation necessitate careful consideration of organizational, technical, and ethical factors. This research provides valuable insights for business leaders and researchers seeking a deeper understanding of the relationship between Artificial Intelligence and business performance.

Langkah-langkah Analisis

1. Preprocessing Teks: Tiga teks digabung ke dalam satu list Python.
2. Ekstraksi Fitur: Menggunakan TfidfVectorizer dari Scikit-learn untuk mengubah teks menjadi representasi numerik berbasis frekuensi kata.
3. Penghitungan Kemiripan: Menggunakan cosine_similarity untuk menghitung kemiripan antar setiap pasangan teks.

Hasil

Notebook mencetak matriks kemiripan (cosine similarity) antara masing-masing teks. Nilai kemiripan berkisar antara 0 (tidak mirip) hingga 1 (sangat mirip).

Souce kode :

https://drive.google.com/file/d/1xHM_Ej0MTMFOTWxvt7VHRcYhKmR9VkZY/view?usp=sharing